

A MAJOR PROJECT
on
“ETHNOBOTANICAL STUDY OF *WITHANIA*
***SOMNIFERA*(ASHWAGANDHA)USED BY TRIBAL**
PEOPLE:PROFILING AND BIOCTIVITY EVALUATION”

Submitted by partial fulfilment for the degree of

MASTER OF SCIENCE IN BOTANY

To

Matoshri Ashatai Kunawar Art's, Commerce & Science Women's College, Hinganghat

By

Miss. Megha Dnyaneshwar Ghode

Under the Supervision of

Dr. Nayana S. Shirbhate

Head Department of Botany

Vidya Vikas Art's, Commerce & Science College, Samudrapur, Dist. Wardha-442301



Department of Botany,

Matoshri Ashatai Kunawar Art's, Commerce & Science Women's College, Hinganghat, Dist.

Wardha- 442301

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VIDYA VIKAS ART'S, COMMERCE & SCIENCE COLLEGE

DEPARTMENT OF BOTANY

SAMUDRAPUR, DIST. WARDHA-442305

Phone: (07151) 225560, 9096943636 Fax: (07151) 225560

Date:

CERTIFICATE

It is certified that I have supervised the work of **Miss.Megha Dnyaneshwar Ghode** on “**Ethnopharmacological Study of *Withania Somnifera*(Ashwagandha)used by Tribal :Profiling and Bioactivity Evaluation**” For the degree of “MASTER OF SCIENCE” in the faculty of Science, Matoshri Ashatai Kunawar Art's, Commerce & Science Women's College, Hinganghat, Dist. Wardha. I find the work is comprehensive and complete in all respect and ready for the submission to the University.

Supervisor:

Dr. Nayana S. Shirbhate

Head department of Botany

Vidya Vikas Art's, Commerce &

Science College, Samudrapur,

Dist. Wardha-442301



Vidya Vikas Shikashan Sanstha, Hinganghat

MATOSHRI ASHATAI KUNAWAR
ARTS, COMMERCE & SCIENCE WOMENS COLLEGE

Principal
Dr. Umesh Tulaskar

Sant Tukadoji Ward, Nandori Road, Hinganghat, Dist. Wardha -442301
(Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

E-Mail ID : makm.college@gmail.com

Mobile No. 9850808807, (Office No. 8483922491) (Estd. 2018)

NAAC B+
ACCREDITED

This is to clarify that **Miss. Megha Dnyaneshwar Ghode** is a Student of M.Sc. Botany (Semester-III) in the Department of Botany, Matoshri Ashatai Kunawar Art's, Commerce & Science Women's college, Hinganghat, Dist. Wardha Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur.

The dissertation in title "**Ethnopharmacological Study of *Withania Somnifera*(Ashwagandha) used by Tribal :Profiling and Bioactivity Evaluation**". Is submitted to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur in partial fulfilment for the degree of Master of Science in Botany.

Forwarded through:

Ms. Darshana S. Rendhe

Head of Department
Matoshri Ashatai Kunawar
Art's, Commerce & Science
Women's College, Hinganghat.

Dr. Umesh P. Tulaskar

Principal
Matoshri Ashatai Kunawar
Art's, Commerce & Science
Women's College, Hinganghat.

DECLARATION

I, Miss. **Megha Dnyaneshwar Ghode** from Post Graduate Department of Botany, Matoshri Ashatai Kunawar Art's, Commerce & Science Women's college, Hinganghat, hereby declare that presented in this dissertation, entitled "**Ethnopharmacological Study of *Withania Somnifera*(Ashwagandha)used by Tribal :Profiling and Bioactivity Evaluation.**" Is an authentic record of my own work for partial fulfilment of the requirement for the awarded degree of "**Master of Science in Botany**". I have worked under the supervision of **Dr. N. S. Shirbhate, Head Department of Botany, Vidya Vikas Art's, Commerce & Science College, Samudrapur.**

Place: Hinganghat.

Miss.Megha Dnyaneshwar Ghode

Date:

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Miss. Megha Dnyaneshwar Ghode

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INTRODUCTION

***WITHANIA SOMNIFERA* (ASHWAGANDHA) USED BY TRIBAL PEOPLE: PHYTOCHEMICAL PROFILING AND BIOACTIVITY EVALUATION.**

INTRODUCTION

Ashwagandha (*Withnina somnifera*) is a dryland medicinal plant having potential uses in herbal drug industry. The dried roots are the primary economic part widely used as sedative, general stimulant, tonic; for treating senile debility, inflammation, rheumatism, female disorders, dropsy, ulcers, scabies, dyspepsia, lumbar pains, abortion and hiccups. Ashwagandha root texture determined by the starch and fibre content plays a significant role in the market price of the roots benefiting farming community. Brittle roots having high starch and low fibre are highly priced because of their ease in making powder and are quoted to be characteristic root textural features of commercial ashwagandha (Atal and Schwarting, 1962).

Winter cherry (*Withnina somnifera*) also called ashwagandha, belongs to the family Solanaceae, is cultivated in the soils that are unsuited for other crops and requires little care. The plant is known for its varied therapeutic uses in ayurvedic and umami practices in India *Withnina somnifera* commonly known for its mefulness in the treatment of arthritis in combination with other drugs, is also credited to be hyperglycaemic and diuretic tic The pharmacological effect of the roots of *W. somnifera* is attributed to withanolides, a group of steroidal lactones. In animals, report is there about activity of tsarina ascribed to its antioxidant property. Earlier studies have reported the absence of any side effects of *W. somnifera* in animals To corroborate the promising results of animal studies in humans, a study was conducted to mass the hyperglycaemic. Hence present results indicate that *W. somnifera* is a good diuretic as reported and that it is highly effective in opening up the natural pores of the body, by enhancing diuresis. Radhika and Andallu *et al.*, (2000)

A mounting body of research is hoisting the credibility of traditional knowledge in meeting the challenges of primary healthcare services and natural resource management globally (Ragupathy and Newmaster, 2009). However, societal domination of western medicine and lack of due recognition to the stakeholders of biological resources, place and value of traditional knowledge base has been impeding the entry of traditional medicine into mainstream healthcare services and its related field of human welfare such as nutrition, environmental assessment and natural resource management (Usher, 2000).

Wound healing process holds several steps which involve coagulation, inflammation, formation of granulation tissue, matrix formation, remodeling of connective tissue, collagenisation and acquisition of wound strength (Suresh Reddy *et al.*, 2002).

In other words wound is a break in the epithelial integrity of the skin and may be accompanied by disruption of the structure and function of underlying normal tissue and may also result from a contusion, hematomas, laceration or an abrasion (Enoch and John Leaper, 2005).

Healing of wounds starts from the moment of injury and can continue for varying periods of time depending on the extent of wounding and the process can be broadly categorized into three. In the Ayurvedic medical system a drug has four key properties, namely rasa, guna, veerya, and vipaka. Rasa constitutes the taste of the drug, Guna means the properties and effects it has on the body following consumption. Veerya represents potency of the drug, whether it has a catabolic or anabolic effect on the body and vipaka means post digestive effect on metabolism. Based up on these factors, mechanism of a drug inside a biological system can be understood in terms of doshas (bio forces that control the body of an organism, whose balance and imbalance regulates health and ill health Senthil Kumar *et al.*, 2006).

A discovery of the healing power of plants through trial and error. Although some of the therapeutic properties attributed to plants have proven to be erroneous, medicinal plant therapy is based on the empirical findings of hundreds and thousands of years. Wound infection is one of the most common diseases in developing countries because of poor hygienic conditions. Wounds are the physical injuries that result in an opening or breaking of the skin and appropriate method for healing of wounds is essential for the restoration of disrupted anatomical continuity and disturbed functional status of the skin (Meenakshi *et al.*, 2006).

Research on wound healing agents is one of the developing areas in modern biomedical sciences and many traditional practitioners across the world particularly in countries like India and China have valuable information of many lesser-known hitherto unknown wild plants for treating wounds and burns. Traditional forms of medicine practiced for centuries in Africa and Asia are being scientifically investigated for their potential in the treatment of wounds related disorders. Ashwagandha (*Withnina somnifera*) is a revered medicinal herb in Ayurvedic tradition, known as a Rasayana (rejuvenator) for promoting vitality, stress resilience, and mental clarity, derived from a shrub native to India, Africa, and the Middle East. Valued for its Adaptogenic properties, it helps the body manage stress and contains withholds, active compounds linked to antioxidant and anti-inflammatory effects, used for anxiety, sleep, and overall well-being, though more research is needed for many claims., (Kumar *et al.*, 2007)

The use of Ashwagandha for its medicinal properties was preached by Punarvasu Atreya around 4000 years ago. Ayurveda described Ashwagandha as a rasayana herb, that is, an herb which has anti-ageing activity and can restore youthfulness by increasing physical strength and immunity. Rasayana also helps in improving memory and intelligence and relieves stress (Kulkarni and Dhir 2008).

Withnina somnifera, commonly known as Ashwagandha, is an important medicinal plant that has been used in Ayurvedic and activity, focusing on two novel activities, tumor inhibition and antiangiogenic properties of withaferin A and the effects of most recent attempts in biotechnological

production of withanolides are also discussed. Indigenous medicine for over 3,000 years. In view of its varied therapeutic potential, it has also been the subject of considerable modern scientific attention. The major Chemical constituents of the *Withnina* genus, chemical constituents of the *Withnina* genus, The withanolides, are a group of naturally The withanolides, are a group of naturally Occurring C28-steroidal lactone triterpenoid built on an intact or rearranged ergostane Framework, in which C-22 and C-26 are Appropriately oxidized to form a six-Membered lactone ring. In recent years, occurring C28-steroidal lactone triterperod membered lactone ring. In recent years, numerous pharmacological investigations have been carried out into the components of *W. somnifera* extracts. We present here an overview of the chemical structures of triterpenoid components and their biological withanolide A on Alzheimer's disease. (Mirjalili *et al.*,2009).

In spite of the commercial significance, root textural parameters have not been explored much. medicinal properties of ashwagandha have been studied well and reviewed in detail Reports on accumulation pattern of total alkaloids accumulation. The pattern of starch and fiber accumulation varied with different crop growth periods. In general, crude fiber content increased with the crop duration, whereas starch content and starch-fiber ratio followed a trend of decrease-increase-decrease as influenced by different phosphates. Discussions on suitable harvest stage based on dry root yield and starch-fiber ratio are made. therapeutically. The root textural quality determined by high starch-fiber ratio fetches premium price for the produce. Influence of morphological variants and crop growth intervals on accumulation of starch and crude fiber in roots was studied to assess morphotypical differences and to determine the right harvest stage

for high root yield with better physical quality. The morphotypes showed similar trend for dry root yield, whereas differences were RR Kumar *et al.*,(2011).

Withnina somnifera is a very significant plant used in a number of orthodox medicinal systems like Ayurveda, Siddha and Unani. Ashwagandha, common name of *W. somnifera*, finds its mention both in the Ayurvedic and Siddha Pharmacopoeia of India (Umadevi *et al.*,2012).

The renowned Indian literatures like Charaka Samhita and Sushruta Samhita have appreciated the importance of Ashwagandha for treating emaciation, rheumatoid arthritis-related inflammations and many other ailments. Ashwagandha is commonly used in India as a home therapy for alleviating hand and limb shivering of elderly people. It is also a very potent aphrodisiac. It is used to prepare a number of medicinal preparations, either alone or in combination with other herbs, for various types of nervous disorders as well. The leaves of Ashwagandha are useful in treating all types of skin lesions, boils, swelling, ulcers, pus formation and inflammation . Umadevi *et al.*, 2012).

However, there are no reports on the contain steroidal compounds including ergostane typeface of carbohydrates or elicitors such as chitosan, steroidal lactones like withaferin A, withanolidesjasmonic acid (JA), acetyl salicylic acid (ASA) and A-Y, withasomniferin A, withasomidicnone, signaling molecules such as nitric oxide on the withasomniferols A-C, withanone, a few flavonoidsaccumulation of withanolides in hairy roots of and reducing sugars. But, low yields from the natural, *w.somnifera*. This communication reports sources, large genotypic and

chemotypes variations accumulation of withanolides in roots, shoots and severe insect attacks during active growth, longeaves collected from untransformed plants and also gestation period between planting and harvesting a enhancement in withaferin A and withanolide A, in some of the constraints for the large scale harvestistably transformed hairy roots of *W. somnifera* as and production of withanolides. In this regard, hairyinfluenced by different carbohydrates, elicitors root cultures have been thought as comparatively fastid a fungicide. Growing and genetically more stable. Madhavi Doma *et al.*,(2012).

W. somnifera root, as used traditionally Alzheimer's disease (AD), is the most common form of senile dementia, affecting more than 15 million people worldwide .With increased life expectancy this number will certainly rise rapidly in the future. AD is characterized by progressive dysfunction of memory and higher cognitive functions associated with memory loss and language deficit which are often accompanied by behavioral and psychological symptoms such as depression, stress, anxiety and mood disturbances The pathological hallmarks are complex and include neuronal degeneration (cholinergic neurons in particular), abnormal neurofibrillary tangles, toxic β -amyloid (AB) plaques, decline of neurochemicals which are essential for neuronal transmission and neuro-inflammation The β -amyloid cytotoxicity to neuronal cells has been identified as one of the major features in AD pathology, but the exact mechanisms involved leading to neurotoxicity still remain an enigma The transmembrane protein CD33 is a sialic acid-binding immunoglobulin-like lectin that regulates innate immunity but has no known functions in the brain, is considered as a risk factor for Alzheimer's disease (AD). Very recently an increased expression of CD33 in (Kurapati *et al.*,2013).

Withania somnifera belongs to the Solanaceae family, commonly known as Ashwagandha, Indian ginseng or Winter cherry. It is an important commercial medicinal crop, considered as similar to *Panax ginseng* in Chinese medicine. Different parts of this plant are used in traditional medicine for the treatment of various ailments. It provides defense against diseases, adverse environmental factors and helps to retard the aging process. Ashwagandha exhibits a wide range of therapeutic properties by tuning the endocrine, cardiopulmonary, central nervous system and sexual behavior without any toxicity. The root has been used most frequently for therapeutic uses and is a constituent of over 200 formulations in Ayurveda, Siddha and Unani medicines. There are several reports to establish its immunomodulatory, anti-inflammatory, antistress, memory enhancing, antiparkinsonian, hypolipidaemic, antibacterial, cardiovascular, antioxidant, antitumor and adaptogenic properties. These properties stem from the characteristic bioactive phytochemicals such as alkaloids and phytosterols present in the plant. This review presents a detailed survey on various traditional uses, phytochemical composition and pharmacological properties of *Withania* Jinu John,(2014)

Experiments using animal models indicated cardio-protective, immunomodulatory, anti-diabetic and neuroprotective effects. Experiments using animal models indicated cardio-protective, immunomodulatory, anti-diabetic and neuroprotective effects of Ashwagandha root extracts. Clinical studies conducted in various conditions suggest that Ashwagandha root extract improves sexual performances in both males and females, aids in reducing and managing stress and anxiety, improves memory and cognition in healthy adults and patients with bipolar disorder. This multipurpose herb also increases muscle strength, muscle size and supports in muscle recovery. In the present study, we have investigated the impact of Ashwagandha root extract powder compared to placebo on insomnia patients as a natural solution to the burgeoning problem of insomnia. Thakur *et al.*, (2015).

Withnina as "Asgand" also finds its mention in the Unani medicine system in Kitab-al-Hashaish. Written by Dioscorides in 78 A.D. *W. somnifera* has found its mention as an official drug in Indian Pharmacopoeia-1985. It is the roots of the plant that have been extensively used in Unani and Ayurvedic systems of medicine. Roots are reported to contain 0.13-0.31% alkaloids, volatile oils, starch, amino acids, glycosides, reducing sugars, and steroids [199]. Besides, the root, leaves are useful to cure fever and swellings, flowers have been used as astringent, diuretics, fruits are used to treat skin ulcers, tumors, carbuncles and seeds are effective in increasing sperm count as well as in testicular growth. This review encompasses the various pharmacological activities of *W. somnifera* (WS) that have been reported in the last two decades with notes on its traditional use (WS) that have been reported in the last two decades with notes on its traditional use, phytochemistry, clinical and toxicological carbuncle, anti-stress, tonic, narcotic, diuretic and against worms, piles, goitre, leucoderma, nervous breakdown, insomnia and constipation. *Withnina* as a medicine system. by NJ Dar *et al.*, (2015).

In Ayurveda *Withnina somnifera* (Ashwagandha) is considered a revered rasayana herb. Rasayana in early ayurvedic medicine means the science of lengthening lifespan. Different parts (leaves, stem, flower, root, seeds, bark and even whole plant) of *Withnina somnifera* have been recommended as an aphrodisiac, liver tonic, anti-inflamatory agent, astringent, to treat bronchitis, asthma, ulcers, mastication, insomnia, and senile dementia etc. The therapeutic use of Ashwagandha for brain related disorders like anxiety, cognitive and neurological disorders and Parkinson's disease is supported by clinical trials and pre-clinical research. Ashwagandha is also used therapeutically as an adaptogen in patients that suffer from nervous exhaustion, insomnia, strums, and as an immune stimulant in patients with low white blood cell counts in blood. Plant derived agents have been used in animal (in vivo) and cell line based models (in vitro) of AD and these have shown multifunctional properties that include pro-cholinergic, anti-oxidant, anti-amyloid and anti-inflammatory activities. Pratte *et al.*, (2018)

The raises in *W. somnifera* market demands in Africa and elsewhere mostly associate with its extensive medicinal, nutritional, and cosmetic applications. In particular, its uses in the productions of pharmaceutical drugs and products for human consumption, including energy drinks, capsules, health supplements, energy boosters, and food product materials, have been remarkably notable Jayanta et al.,(2018)

In recent times, Insomnia or sleeplessness has become a common disorder that is affecting a large global population and impairing the general health and mental wellbeing. Insomnia is clinically characterized by difficulty in sleep onset or sleep maintenance or a con both and impairment of dailywellbeing. Insomnia is clinically functionalities. The clinical criteria that designate insomnia are average sleep latency of more than 30 min, wakefulness after sleep onset of more than 30 min, sleep efficiency of less than 85% of total sleep time of fewer than 6.5 hours Insomnia often leads to fatigue, energy depletion, impairment in concentration and increased irritability. Investigations of etiological and pathophysiological factors since decades suggest that the onset and the progress of insomnia depend on multiple factors including genetic, cognitive and behavioral, physiological, biochemical, and environmental. Deepak Langade *et al.*,(2019)

Withania somnifera is a very unique medicinal plant explored by number of orthodox medicinal systems such as Ayurveda, Siddha and Unani. It is commonly known as Ashwagandha, Indian ginseng and winter cherry, and finds its potential medicinal properties in the Ayurvedic Pharmacopoeia of India and Siddha Pharmacopoeia of India. The Withania genus is classified under the Solanaceae family and includes around 60 species, among which *W. somnifera* and *W. coagulans* are often mentioned in Ayurveda. Ashwagandha is a treasure house of widespread array of metabolites such as steroids, flavones, alkaloids, carbohydrates, glycosides, saponins, tannins, trapezoids and Coumadin. Eight different polyphenols (five phenolic acids, vanillin, benzoic, p-coumaric, garlic and syringic acid, and three types of flavonoids, naringenin, catechin and kaempferol) were reported from Ashwagandha. Each part of this plant holds an assortment of different metabolites, and the metabolite concentrations vary among the different chemotypes. Ashwagandha has been used for a variety of ailment since a long time by traditional medicinal systems. It has been used for a range of diseases such as diabetes, emaciation, arthritis and rheumatoid arthritis-related inflammations, some kinds of seizures, diarrhoea, dermatitis and insect bite and specially used in the treatment of nervous disorder. The importance of Ashwagandha in medicinal treatments has attracted the attention of a large number of scientists; as a result, numerous experiments have been carried out, which have verified the therapeutic properties of Ashwagandha. P. poojari *et.al.*,(2019).

In the current world healthcare scenario, more than 80% of the world population, particularly in developing countries, trusts phytomedicines for the therapies needs (1). Still prevailing success of traditional systems of therapies such as Ayurveda, Greco-Arab (Unani-Tibb), and Chinese medicine

system is due to the fact that phytomedicines have fewer side effects and are economical. The plants belonging to the genus *Withania* of Solanaceae family are generally acclaimed as medicinally essential due to the high therapeutic and nutraceutical potential. One of the plants of the genus, *W. Somnifera*, possessed a plethora of medicinal uses and pharmacological applications. *W. somnifera* (Syn: ashwagandha, suranjan, winter cherry, Indian ginseng) is a xerophytic plant that nurtures abundantly in Africa, the Mediterranean, Sri Lanka, Pakistan, and India. In the Ayurvedic system of medicines, roots and leaves of the plant were considered phototherapeutic agents to cure various ailments. Various clinical and preclinical trials exhibited the plant's potential in curing hepatotoxicity, neurological disorders, anxiety, Parkinson's disease, and hyperlipidemia. The fruits contained considerable amounts of saponins and leaves possessed insect repellent properties. Phytochemical analysis of *W. somnifera* revealed the presence of pharmacologically active steroidal lactones named withanolides. Withanine, a group of alkaloids isolated from the roots of the plant, forms 38% of the total weight of alkaloids. S Saleem *et al.*, (2020)

W. somnifera is extensively traded and used as a medicinal herb in India due to its potentially active compounds. Ashwagandha is reported to increase the human's capacity to manage pressure and cell-mediated resistance, along with fortifying balance in the immune system (Zahiruddin *et al.*, 2020).

, the global prevalence of mental health issues and sleep disturbances has increased, leading healthcare providers and the public to seek integrative methods beyond traditional drug therapies. As worries regarding the side effects and shortcomings of usual treatments for mood disorders, anxiety, and insomnia have grown, the quest for effective, evidence-based natural remedies has intensified. One plant that has garnered notable scientific and clinical attention is *Withania somnifera*, also known as ashwagandha. Ashwagandha is a small shrub native to the Indian subcontinent, the Middle East, and parts of Africa. The review primarily included in vivo studies, particularly clinical trials. Through synthesizing the available evidence, we aim to determine whether ashwagandha can serve as a viable adjunct or alternative to standard treatments, while also identifying areas where further research is needed. Hk Afewerjy *et al.*, (2021).

The primary bioactive constituents present in *W. somnifera* are the steroidal lactones known as withanolides (withaferin A and withanolide D), which target different biomolecules in the living systems and are responsible for their diverse pharmacological activities (Paul *et al.*, 2021)

Studies have suggested that Ashwagandha may exhibit cardioprotective properties, be helpful in the treatment of sleep disorders, improve stress resilience, reduce anxiety, be helpful in hypothyroidism, and enhance muscle strength and recovery. While the potential health benefits of Ashwagandha are promising, further research is needed to fully understand its mechanisms of action, attempting to break down and remove damaged and dead neurons as well as senile plaques. *Alzheimer's* cells produce toxins, destroying consisting of β -amyloid. Physiologically, τ proteins stabilize

microtubules, along with other proteins. Accumulating senile plaques are accompanied by microglia (inflammatory response cells), attempting to break down and remove damaged and dead neurons as well as senile plaques. L1a cells produce toxins, destroying consisting of β -amyloid. Physiologically, τ proteins stabilize microtubules, along with other proteins. Accumulating senile plaques are accompanied by microglia (inflammatory response cells), attempting to break consisting of β -amyloid. Physiologically, τ proteins stabilize microtubules, along with other proteins. Accumulating senile plaques are accompanied by microglia (inflammatory response cells), attempting to break down and remove damaged and dead neurons as well as senile plaques. L1a cells produce toxins, destroying both @ed and healthy cells and enhancing the brain's inflammatory response P Mikulska *et al.*, (2023).

The mechanism of drug use by *W. somnifera* has not been elucidated. Accordingly, there are unreasonable points in the current *W. somnifera* quality management system. In fact, its medicinal value has received less attention, despite its long history of clinical application. In this review, information regarding various aspects of *W. somnifera* was collected from peer-reviewed journals for the period 2000-2022. PubMed, Shod Ganga, Web of Science, Google Scholar, etc., was used to retrieve electronic information. Based on this information, we performed a full review of *W. somnifera* aiming to provide information to better understand the pharmacological mechanism and potential medicinal applications of this species, and to develop a reasonable quality control system for *W. somnifera* as medicinal material in the future. H. Gaurav *et al.*, (2023).

In the past, many communities approached traditional medicine systems in order to treat any physiological disorder by using plant products with their unique formulas. However, without any established scientific evidence and safety trials, these herbs proved effective in treating respective diseases. Recent studies have proved the presence of bioactive ingredients responsible for their therapeutic potential (Al-Harrasi *et al.*, 2023).

According to the World Health Organization (WHO), approximately 80% of the population in developing countries rely on traditional remedies, predominantly plant-based, for primary healthcare needs This widespread usage not only sustains cultural practices but also supports rural economies and indigenous knowledge systems through the cultivation, trade, and local utilization of botanicals. Moreover, the growing consumer preference for natural health products has positioned medicinal plants as a multibillion- dollar with increasing demand worldwide. In In parallel, medicinal plants are emerging as valuable tools in addressing major global health challenges. The alarming rise of antimicrobial resistance (AMR) has prompted a resurgence of interest in phytochemicals with novel antimicrobial properties, which may complement or substitute synthetic antibiotics. Similarly, as non-communicable diseases (NCDs) such as diabetes and cardiovascular disease become increasingly prevalent, bioactive plant compounds offer promising preventive and therapeutic benefits with fewer adverse effects compared to conventional drugs. However, the widespread exploitation of these plant resources poses risks to biodiversity and environmental sustainability, particularly where harvesting is unregulated or destructive., adherence to

international conservation frameworks are urgently required to safeguard these resources for future generations Al Raish SM, *et al.*,2025)

In recent years, the global prevalence of mental health issues and sleep disturbances has increased, leading healthcare providers and the public to seek integrative methods beyond traditional drug therapies. As worries regarding the side effects and shortcomings of usual treatments for mood disorders, anxiety, and insomnia have grown, the quest for effective, evidence-based natural remedies has intensified. One plant that has garnered notable scientific and clinical attention is *Withania somnifera*, also known as ashwagandha. health issues and sleep disturbances has increased, leading healthcare providers and the public to seek integrative methods beyond traditional drug therapies. The quest effective, evidence-based natural remedies has intensified Michał Wiciński et al.(2025).

REVIEW OF LITERATURE

Radhika and Andallu *et al.*, (2000) evaluated the hyperglycaemic, diuretic, and hypocholesterolemic effects of powdered roots of *Withnina somnifera* in human subjects. The study involved six patients with mild non-insulin- dependent diabetes mellitus (NIDDM) and six patients with mild hypercholesterolemia, who were administered dietary patterns were monitored throughout the study. The study confirmed that the root of *W. somnifera* is a potential source of hyperglycaemic, diuretic, and hypocholesterolemic agents. It helped establish that Ashwagandha roots possess definite therapeutic potential in humans, not just in animal models. This study is frequently cited in modern reviews discussing the metabolic syndrome management potential of *withnina somnifera*. This study demonstrated the potent “Rasayana” (rejuvenating) properties described in Ayurvedic medicine, which aims to improve metabolic and renal function while reducing cardiovascular risk factors.

Monty (Mohanty) I.R., Arya D.S., Gupta S.K. *et al.*, (2008). *Withnina somnifera* provides cardioprotection and attenuates ischemia-reperfusion induced apoptosis. Clinical Nutrition. Ischemia-reperfusion (I/R) injury is a major cause of myocardial damage, primarily mediated by oxidative stress, calcium overload, and apoptosis. Medicinal plants with antioxidant properties have gained attention as potential cardioprotective agents. *Withnina somnifera* (Ashwagandha), a well-known Ayurvedic herb, has documented antioxidant and adaptogenic properties, but its role in myocardial I/R injury required further investigation.

Mohammad Hossein Mirjalili *et al.*, 2009. *Withnina somnifera*, commonly known as Ashwagandha, is an important medicinal plant that has been used in Ayurvedic and indigenous medicine for over 3,000 years. In view of its varied therapeutic potential, it has also been the subject of considerable modern scientific attention. The major chemical constituents of the *Withnina* genus, the withanolides, are a group of naturally occurring C28- steroidal lactone triterpenoids built on an intact or rearranged ergostane framework, in which C-22 and C-26 are appropriately oxidized to form a six-membered lactone ring. In recent years, numerous pharmacological investigations have been carried out into the components of *W. somnifera* extracts. We present here an overview concentration of spermatozoa and the metal ions copper, zinc, iron and gold in seminal plasma from the chemical structures of triterpenoid components and their biological activity, focusing on two novel activities, tumor inhibition and antiangiogenic properties of withaferin A and the effects of withanolide A on Alzheimer's disease. The most recent attempts in biotechnological production of withanolides are also discussed.

Mahdi A.A. *et al.*, (2011) *Withnina somnifera* Improves Semen Quality in Stress-Related Male Fertility “Stress has been reported to be a causative factor for male infertility. *Withnina somnifera* has been documented in Ayurveda and Unani medicine system for its stress-combating properties.

However, limited scientific literature is available on this aspect of *W. somnifera*. We undertook the present study to understand the role of stress in male infertility, and to test the ability of *W. somnifera* to combat stress and treat male infertility. We selected normozoospermic but infertile

Shukla KK, *et al.*, (2011) This study investigated the effect of a 3-Month treatment with *Withania somnifera* on apoptosis and intracellular reactive oxygen species (ROS) in infertile men. Concentration of spermatozoa: Oligozoospermic, (normozoospermic, $n = 25$ oligozoospermic, $n = 25$ and asthenozoospermic, $n = 25$). The apoptotic and necrotic cell distribution were analysed by annexin-V binding and propidium iodide uptake using flow cytometry. ROS generation was measured by fluorescence intensity and metal ions were analysed by atomic absorption spectrophotometry. The results demonstrated that, prior to treatment, sperm apoptosis and intracellular ROS concentrations.

Madhavi Doma *et al.*, (2012) Leaves of *Withania somnifera* contained more withaferin A and withanolide A than roots indicating that these compounds mainly accumulate in leaves. With an increase in age of the plant, withaferin A was enhanced with a corresponding decrease in withanolide A. Hairy root cultures were induced from leaf explants using *Agrobacterium rhizogenes* and the transgenic nature of hairy roots was confirmed by partial isolation and sequencing of gene, which could not be amplified in untransformed plant parts. In hairy roots, withaferin A accumulated at 2, 3 and 4% but not at 6% sucrose, the highest 4% level. High and equal amounts of (890 and 886 $\mu\text{g/g}$ dry tissue concentrations of glucose enhanced, Plant in India and in other regions. Withanolides,” mainly in leaves and Roots. Hence, Govt. of India has graded this Aromatic plant in top 36 cultivated Precious medicinal plants in focus of immense requirement for market and Medicine production. The traditional Cultivation method of *W. somnifera* suffers from severe shortage in production to provide sufficient raw materials for Phytochemical synthesis and medicinal Formulations due to biotic and abiotic Factors, poor seed germination, loss or low Seed viability, etc.

Soman *et al.*, (2012) Oral administration of *W. somnifera* root extract and withanolide-A at doses of 100 mg/kg/day and 10 mol/kg/day for 15 days restored spatial memory deficit in male Wistar rats with epilepsy induced by pilocarpine via inhibiting oxidative stress-induced alteration in glutamergic transmission in the brain by reducing N-methyl-D-aspartate (NMDA) receptor expression. Furthermore, in a porcupine-induced epilepsy model, suppressing the α -Amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA receptor) expression improved motor learning.

BM, Shilpa BD, Laxmipriya D, and Modi *et al.*, (2012), is a randomized open clinical trial evaluating a combined Ayurvedic treatment for menopausal syndrome. The study found that this combination is a safe, effective, and significant alternative to modern therapies, such as Hormone Replacement Therapy (HRT), for managing somatic and psychological complaints in women with

mild to moderate symptoms. Study Design & Participants: The trial registered 52 patients, with 51 completing the study, aimed at evaluating the clinical efficacy of a 12-week intervention. Significant Symptom Reduction.

Kurapati K.R.V., *et al.*,(2013) Alzheimer's disease (AD) is characterized by progressive dysfunction of memory and higher cognitive functions with abnormal accumulation of extracellular amyloid plaques and intracellular neurofibrillary tangles throughout cortical and limbic brain regions. At present no curative treatment is available, and research focuses on drugs for slowing disease progression or providing prophylaxis. *Withnina somnifera* (WS) also known as 'ashwagandha' is used widely in Ayurvedic medicine as a nerve tonic and memory enhancer. However, there is a paucity of data on the potential neuroprotective effects of *W. somnifera* against β -Amyloid (1-42)-induced neuropathogenesis. At the minimum inhibitory concentration (MIC) values further confirmed the antimicrobial potential of the plant extracts, with the lowest MIC of 6.25 mg/ml against *S. typhi* and 12.5 mg/ml against *E. coli* these findings demonstrate that *W. somnifera* possesses. Introduction: The most prevalent neurodegenerative illness and kind of dementia is Alzheimer's disease (AD). It shows up as a reduction in short-term memory and cognitive function that affects day-to-day functioning. Identification, which when combined with neuropathology provides crucial hints regarding the broader causes.

Grover *et al.*,(2013) *Withnina somnifera* well-known medicinal plant widely used in traditional systems like Ayurveda due to its diverse therapeutic properties. The pharmacological activity of this plant is mainly attributed to withanolides, a group of steroidal lactones with significant medicinal value, including anti-inflammatory, anti-cancer, and anti-stress effects. However, the natural production of withanolides in plants is relatively low and often insufficient to meet growing commercial demand. In this context, metabolic engineering approaches have gained importance to enhance secondary metabolite production. One key enzyme in the withanolide biosynthetic pathway is Squalene synthase, which catalyzes an essential step in sterol and triterpenoid formation. The study by Grover A and colleagues (2013) focuses on increasing withanolide content by overexpressing the squalene synthase gene in *Withania somnifera*, providing .

Siriwardane *et al.*,(2013) Ethanolic extract of the *W. somnifera* stem bark mainly contains withanolides, including somniferanolide, somniwithanolide, somnifera withanolide, withasomnilide, and withasomniferanolide .

Pratte *et al.*,(2014) found that all five analyzed human trials, utilizing *Withnina somnifera* (Ashwagandha) as an intervention for anxiety, reported significantly greater improvements in anxiety scores compared to placebos, Highlighting its potential as a safe and effective botanical treatment for reducing anxiety and stress. Search terms included "ashwagandha," "Withnina

somnifera,” and terms related to anxiety and stress.

Baitharu et al.,(2014)The root extract of *W. somnifera* has also been shown to mitigate memory loss in male rats by increasing reduced glutathione (GSH) levels through activation of glutathione biosynthesis in hippocampal cells showing some benefit in Alzheimer’s disease .

Kumar et al.,(2014) The neuroprotective potential of *W. somnifera* has also been proved by exposing C6 cells to lead nitrite by balancing glial fibrillary acidic protein (GFAP) expression, heat shock protein (HSP70), and neural cell adhesion molecule (NCAM) expression, the unbalance of the same results in neurodegeneration .

Bakhtiar Choudhary et al., (2015) .Ashwagandha (*Withnina Somnifera* has been traditionally used for various actions ranging from vitalize, improve endurance and stamina, promote longevity, improve immunity, and male and female fertility. However, clinical studies are needed to prove the clinical efficacy of this herb, especially in cardiovascular endurance and physical performance. Cardiorespiratory Endurance was assessed by measuring the oxygen consumption at peak physical exertion (VO₂ max) levels during a 20 m shuttle run test. The World Health Organization self-reported QOL questionnaire (physical health, psychological health, social relationships, and environmental factors) was used to assess the QOL. Student’s t-test was used to compare the differences in a mean and change from baseline VO₂ max levels, whereas Wilcoxon signed-rank test was used to assesses.

Pandey et al.,(2018)demonstrates that withanone, a key phytochemical derived from *Withnina somnifera* (*Ashwagandha*) roots, provides multifunctional neuroprotective effects against cognitive dysfunction. It reverses cognitive deficits by inhibiting amyloid-beta aggregation, reducing oxidative stress, and mitigating neuroinflammation, showing potential for Alzheimer’s disease treatment. Penconazole (PEN) is a systemic triazolefungicide used to control various fungal many hazardous effects on population including neurotoxicity. *Withnina* and neurotoxicity was evaluated, and diseases on grapes, stone fruits, cucurbits,and strawberries. Still, it leaves residueson treated crops after collection withasomniferin leaves extract (WSLE) is known for its memory and brain functionenhancing ability. To evoke such action efficiently, WSLE bioactive metabolitesare needed to cross the blood-brainbarrier, that could limit the availability of such compounds to be localized withinthe brain. Therefore, in the present study the association between PEN exposure formulated WSLE nanoemulsion wasinvestigated for improving thepermeability of the plant extract acrosspermeability of the plant extract across the blood-brain barrier.

Jayanta *et al.*,(2018) The raises in *W. somnifera* market demands in Africa and elsewhere mostly associate with its extensive medicinal, nutritional, and cosmetic applications. In particular,

its uses in the productions of pharmaceutical drugs and products for human consumption, including energy drinks, capsules, health supplements, energy boosters, and food product materials, have been remarkably notables.

Deepak Langade *et al.*, (2019) Ashwagandha root extract is an effective and safe natural compound for improving sleep quality and managing anxiety in patients with insomnia. Literature suggests Ashwagandha effects are more prominent in adults diagnosed with insomnia and when used at dosages ≥ 600 mg/day for at least 8 week. Recent research indicates that active compounds in Ashwagandha, such as triethylene glycol (TEG), may be responsible for its sleep-inducing potential by interacting with GABAA receptors. Beyond sleep, other reviews confirm Ashwagandha role as an adaptogen that reduces serum cortisol levels and helps the body manage physical and psychological +3Stress.

Deshpande *et al.*, (2020), Sleep disturbances, particularly non-restorative sleep (NRS), affect a significant proportion of the population and are associated with fatigue, reduced cognitive function, and poor quality of life. Traditional Ayurvedic medicine describes Ashwagandha (*Withnina somnifera*) as a “Rasayana” herb with adaptogenic and sleep-inducing properties. Modern research has increasingly explored its role in improving sleep physiology and stress response. Objective: Non-restorative sleep (NRS) affects 10% people worldwide, leading to poor sleep quality, as well as physical and cognitive fatigue. This is the first human study in which an extract of ashwagandha (*Withnina somnifera* Dunal L.) was evaluated for effects in improving overall sleep quality in subjects with NRS

Vikrant Mehta *et al.*, (2021) Increasing herbal formulations have been used to treat several diseases including cancer. *W. somnifera* (Ashwagandha) is one such plant the extracts of which have been tested against a number of ailments including cancer, which remains as one of the most dreadful diseases on the globe. The ever-increasing number of cancer related mortality demands the development of novel chemopreventive agents with minimum side effects. Different compounds isolated from various parts of the plant like root, stem, and leaves have been reported to display significant anti-cancerous and be used alone or in combination with other chemotherapeutic drugs for cancer treatment. Through this review, we highlight the importance of *W. somnifera* in countering the potential oncogenic signaling mediators that are modulated by active constituents of *W. somnifera* in a variety of cancer types.

D Pandya *et al.*, (2022) Solid tumors account for ~60% of podiatric, as well as adolescent and young adult (AYA), cancers, and outcomes for patients with these progressive diseases remain poor. This highlights the critical need to develop tumors models from patients with aggressive cancers so that oncogenic signatures can be identified for therapeutic testing. Thus, patient-derived xenografts (PDXs) were established from sarcoma and Wilms tumor patients at diagnosis or following treatment. Overall, the molecular landscape of serially passaged PDXs recapitulated the

original tumors based on an integrated multi-OMICS pipeline that cross-validated cancer- associated pathways. Actionable mechanisms of tumor progression (PDXs) were established from sarcoma and Wilms tumour patients at diagnosis or following treatment. Overall, the molecular landscape of serially passaged PDXs recapitulated the original based on an integrated multi-OMICS were identified. CDK4/6 and BETS were prioritized as biomarkers of therapeutic response for in vivo validation

BP Mikulska *et al.*, 2023). Indian ginseng is also known as Indian winter cherry, Ashwagandha, or the herb *Vitania sluggard (Withnina somnifera)*. The raw material used in medicine is the root, and the name “Ashwagandha” is derived from the word “ashwa”, meaning horse. It is believed that after consuming the root, one gains powers similar to that of a horse. The second part of the name “gandha,” means fragrance and refers to the characteristic smell of the fresh root of the plant]. Since ancient times, it has been traditionally used in Ayurvedic medicine as a substance that strengthens the nervous system. This is evidenced by its adaptogenic effects and medicinal uses

Gaurav *et al.*, (2023) *Withnina somnifera* L. Dunal (Ashwagandha), a key medicinal plant native to India, is used globally to manage various ailments. This review focuses on the traditional uses, botany, phytochemistry, and pharmacological advances of its plant-derived constituents. It has been reported that at least 62 crucial and 48 inferior primary and secondary metabolites are present in the *W. somnifera* leaves, and 29 among these found in its roots and leaves are chiefly steroidal compounds, steroidal lactones, alkaloids, amino acids, etc. In addition, the whole shrub parts possess various medicinal activities such as anti-leukotriene, antineoplastic, analgesic, oxidant, immunostimulatory, and of its medical use remains unknown. This review provides a comprehensive as for the development of a rational quality control system for *W. somnifera* as a therapeutic material in the future.

Arjun Singh (2023) medicine is a set of knowledge, abilities, and procedures based on assumptions, beliefs, and experiences of traditional societies in order to preserve their health. In many undeveloped countries, many rural or indigenous people place a great importance on traditional herbal medicines. The World Health Organization estimates that 60% of rural Indians use. The usage of herbal supplements increased from 2.5% to 12% over the previous five years. India’s transition from traditional to modern medicine has been made easier by the examination of novel drugs, especially those made using components derived from plants. Tannins, alkaloids, sugars, terpenoids, steroids, flavonoids, and phenols are a few of the chemical elements included in therapeutic plants that have a specific physiological impact on the human body. Natural therapeutic properties are just one benefit of medicinal plant.

Vivek Basudkar *et al.*,(2024) Inflammaging, a coexistence of inflammation and aging, is a persistent, systemic, low-grade inflammation seen in the geriatric population. Various natural compounds have been greatly explored for their potential role in preventing and treating Inflammaging. *Withnina somnifera* has been used for thousands of years in traditional medicine as a nutraceutical for its numerous health benefits including regenerative and adaptogenic effects. Recent preclinical and clinical studies on the role of *Withnina somnifera* and its active compounds in treating aging, inflammation, and oxidative stress have shown promise for its use in healthy aging. We discuss the chemistry of *Withnina somnifera*, the ethology of inflammaging and the protective role(s) of *Withnina somnifera* in inflammaging in key organ systems including brain, lung, kidney, and liver as well as the mechanistic underpinning of these effects. Furthermore, we elucidate the beneficial effects of *Withnina somnifera* in oxidative stress/DNA damage, immunomodulation, COVID-19, and the microbiome.

Raish SM, *et al.*,(2025) Medicinal plants have long been studied for their therapeutic applications, yet few integrative models exist to conceptualize how these species function across multiple health domains, particularly in the context of metabolic and infectious diseases. A growing body of literature supports the role of phytochemicals such as flavonoids, phenolic, alka-loids, and saponins in modulating oxidative stress, inflammatory and metabolic enzyme pathways Studies have shown that *P. dactylifera*, *S. lycopersicum*. Ashwagandha root extract significantly reduces body weight and Body Mass Index (BMI) compared to a placebo, alongside improving quality of life. Studies noted that ashwagandha is generally safe, with some reporting mild side effects such as nausea or abdominal pain, which often resolve without

PP Jamnekar *et al.*,(2025)Ashwagandha (*Withania somnifera*), well-established herb in Ayurvedic medicine, is increasingly researched for its adaptogenic properties and regulatory roles in neuroimmune processes. The present review aims to integrate mechanistic, preclinical, and clinical evidence on ashwagandha immunomodulatory, neuroprotective, psychiatric, sleep-regulating, and anti-inflammatory activities, with emphasis its bioactive compounds, such as withanolides, sitoindosides, and alkaloids. Such compounds modulate the hypothalamic-pituitary-adrenal (HPA) axis, inhibit NF-kB, induce Nrf2 activation, and affect gamma-aminobutyric acid (GABA)ergic signaling, collectively contributing to its anti-inflammatory, antioxidant, and anxiolytic actions. Clinical trials with standardized ashwagandha extracts have shown reductions in stress-related ashwagandha extracts have shown reductions in stress-related biomarkers, along with improvement in cognitive performance sleep quality and mood parameters Neuroprotective actions have also been demonstrate .

MATERIAL AND METHODS

MATERIAL AND METHODS

Botanical name: *Withnina Somnifera*

Ashwagandha is also known as "Indian ginseng". According to Ayurveda, it is a very useful plant. Its roots are used to cure debility, anemia, and impotence. The regular use of ashwagandha improves vigor and immune power. Ashwagandha Cultivation is a hardy and drought-tolerant plant that grows well in dry regions.

The global interest in this plant and the high demand for its roots provide ample scope to cultivate this plant on commercial scale. Other opportunities for cultivation include: Present price for roots is attractive, crop gives economically remunerative returns in comparison to traditional crops, ease of cultivation under rainfed condition, the crop can be integrated with traditional crops through crop sequencing, opportunities for marketing leaf and seed exist, by-products can be profitably be utilized, value addition can increase profits, however, current exports are limited and large scale exports of roots and value added products need to be explored. In India, it is found in the plains of Madhya Pradesh, Gujarat, Rajasthan, western Uttar Pradesh, Maharashtra, Karnataka, Gurkha Punjab and Haryana and in the plains of Himalayas up to 1200 meters. Ashwagandha is cultivated on a large scale (in about 10 thousand hectares) in Mandsaur and Neemuch regions of Madhya Pradesh. Chemical Organization: At its root flyable oil, steroidal lactones: withanolide and steroidal lactones (withanolides, withaferins) and saponins (Mishra et al., 2000) are found, of which somnifera is the main one.



Fig 1.1

The seeds set the foundation for top-quality herbs when growing any plant. We purchased our original Ashwagandha seeds from Richo Cech, founder of Strictly Medicinal Seeds in Williams, Oregon, which prides itself on providing organic, GMO-free, non-hybrid seeds. A close-up of a pair of hands gently holding rich, dark soil, emphasizing a connection to natural, organic cultivation. The texture of the soil appears moist and fertile, suggesting careful nurturing and sustainable farming practices. Surrounding this visual is a section of a website promoting herbal products, where the accompanying text highlights the importance of high-quality seeds in producing potent plants. It explains that the company sources its original ashwagandha seeds from a reputable organic supplier known for non-GMO, non-hybrid seeds, and has since developed a self-sustaining process by

harvesting and preserving its own seeds annually. Although this method requires more time and effort, it ensures that the crops are well-adapted to the local environment and growing conditions, ultimately leading to better quality herbs.

Tribal communities and traditional medicine systems use different parts of the *Withnina somnifera* (Ashwagandha) plant, particularly the roots and leaves, to treat a variety of ailments, employing methods such as preparing powders, pastes, decoctions, and infusions.



Fig 1.1

Medicinal plants have long been a vital part of healthcare, both in traditional practices and modern medicine. Among these, many plants with spreading roots called rhizomes and tiny seeds stand out for their healing properties. These plants contain natural bioactive compounds that help treat a variety of diseases. This project is dedicated to discovering such plants, exploring their unique structures, and understanding their beneficial properties. By studying how these plants work and their potential applications, we aim to uncover new ways they can support health and wellness. Ultimately, this research not only highlights the importance of these remarkable plants but also bridges ancient wisdom with modern science, offering promising possibilities for future healthcare solutions. Objectives The identification of plant species through roots and seeds is a crucial method in botanical research, especially when the aerial parts of plants are absent or unclear. Roots, particularly rhizomes, exhibit distinctive traits such as colour, texture, branching patterns, and growth direction, which are essential for species differentiation. Similarly, seeds provide key identification markers including size, shape, and surface texture Examining the structure and growth patterns of rhizomes reveals how these plants spread horizontally underground, facilitating rapid vegetative propagation and enhancing survival under challenging environmental conditions.

This growth strategy enables regeneration from small root fragments, contributing significantly to the plant's resilience. Furthermore, analyzing the medicinal and economic significance of rhizomatous plants underscores their importance in both traditional and modern healthcare. Many of these plants contain bioactive compounds with anti-inflammatory, antimicrobial, and digestive benefits. Economically, they find applications in herbal medicine, agriculture, and soil conservation. Understanding their primary ecological roles—such as preventing soil erosion, enhancing soil fertility through organic matter decomposition, and acting as natural ground cover—highlights their broader environmental value. Overall, studying roots and seeds not only facilitates accurate plant identification but also offers valuable insights into their ecological importance, sustainable utilization, and potential applications in medicine and industry This medicinal plant belongs to the Solanaceae family and is widely used in traditional

Ayurvedic medicine. The small, lantern-like husks seen in the hand enclose bright red berries, which contain seeds. While the roots are the most valued part for their adaptogenic properties—helping the body manage stress, improve energy, and support immunity—the fruits and leaves also have medicinal uses. The plant is known for its bioactive compounds called withanolides, which possess anti-inflammatory, antioxidant, and neuroprotective effects. Ashwagandha is commonly used to reduce anxiety, enhance stamina, improve sleep, and support overall health, making it one of the most important herbs in herbal medicine systems.

METHOD

- Field collection of plant samples (roots, stems, seeds)
- Morphological observation (size, color, texture)
- Literature review of medicinal properties



Fig1.3

The plant is propagated using seeds. Seeds can be directly sown in the main field or started in a nursery for transplanting. Seeds are sown in nursery beds just before the rainy season and germinate in 6-7 days. Seedlings are transplanted when they are about 6 weeks old, typically during the late rainy season around August- Seeds. Soil and Climate W. somnifera grows well in sandy loam or light red soil, having With good drainage. It can be cultivated between 600-1200 m altitude. The semi-tropical areas receiving 500-750 mm rainfall are suitable for cultivation of this rained crop. The crop requires dry season during its growing period. Temperature between 200C to 350C is most suitable for cultivation. Late winter rains are conducive for the proper Ashwagandha cultivation. The soil of the field selected for Ashwagandha cultivation is well pulverized by ploughing, disking and/or harrowing. The field may be then levelled by the application pita. Development of the plant roots. It can be cultivated between 600-1200 m altitude. The semi-tropical areas receiving 500-750 mm rainfall are suitable for cultivation of this rained crop. The crop requires dry season during its growing period. Temperature between 200C to 350C is most suitable for cultivation. Late winter rains are conducive for the proper Ashwagandha cultivation. It the soil of the field selected for Ashwagandha cultivation is well pulverized by ploughing, disking and/or harrowing. The field may be then levelled by the application pita.

Development of the plant roots. We start our Ashwagandha seeds in the greenhouse, where

the tiny seedlings sprout within 14 days. After 6-8 weeks, our farmers transplant the delicate seedlings to outdoor raised beds piled high with nutrient-rich soil. We add our farm-made compost to the planting holes along with organic matter and Mountain Microorganisms (MM), a natural organic fertilizer we use to ensure good-quality microbiological activity. With raised beds piled high with nutrient-rich soil. We add our farm-made compost to the planting holes along with organic matter and Mountain Microorganisms (MM), a natural organic fertilizer we use to ensure good-quality microbiological activity. With raised beds and no tilling, the soil has more space for microorganisms to thrive, which is an important part of the regenerative farming process. Start our Ashwagandha seeds in the greenhouse, where the tiny seedlings sprout within 14 days. After 6-8 weeks, our farmers transplant the delicate seedlings to outdoor raised beds piled high with nutrient-rich soil. We add our farm-made compost to the planting holes along with organic matter and Mountain Microorganisms (MM), a natural organic fertilizer we use to ensure good-quality microbiological activity. With raised beds piled high with nutrient-rich soil.

We add our farm-made compost to the planting holes along with organic matter and Mountain Microorganisms (MM), a natural organic fertilizer we use to ensure good-quality microbiological activity. With raised beds and no tilling, the soil has more space for microorganisms to thrive, which is an important part of the regenerative farming process. Growing Ashwagandha in raised beds is also helpful because it helps the roots grow abundantly straight and long without being obstructed by compacted soils. After the Ashwagandha seedlings are planted and established we plant Sunna Hemp *crotalaria juncea* seed, a beautiful yellow flowering plants that provides shade and nitrogen for the Ashwagandha. The Sunna Hemp release ammonium nitrate into the soil, which binds to the Land Preparation Ashwagandha is usually grown in fields which are not well covered by the irrigation systems. The field on which food crops cannot be taken profitably for the above reason may be used for Ashwagandha cultivation. The soil of the field selected for Ashwagandha cultivation is well pulverized by ploughing, disking and/or harrowing. The field may be then levelled by the application pita.

The crop can be sown either by broad casting or in lines. Live to line method is preferred as it in creases root production and also helps in performing intercultural practices properly. The seeds are usually sown about 1-3 cm deep in June July in nursery. A light shower after shower after sowing ensures good germination. About 500-750 gm seeds are sufficient for 1 ha. Field. Seeds can be treated, with Thiram or Indofil or Dithane medicinal plants before sowing to protect seedlings from seed borne diseases. The seedling after 25-35 days after sowing can be transplanted in the field maintaining Spacing between the plants & the rows. It may be noted that since Ashwagandha is a late rainy season Kharif crop, the time of sowing is decided by date of arrival of monsoon in that area. The seeds sown by broadcasting or in the line in furrows should be thinned out by hand at 25-30 days after sowing to maintain a plant population of about 30-60 plants per square meter (about 3.5 to 6 lakh plants/hectare). The plant density to be used may depend on the nature and fertility of the soil. On the marginal land the population is kept high. If some fertiliser is applied then the population should preferably be kept at a lower level. One hand weeding at an

early stage is sufficient to enable the Ashwagandha plants to take over the growth of weed which get suppressed by its smothering effect.

The medicinal plants have to be grown without chemical fertilizers and use of pesticides. Organic manures like, Farm Yard Manure (FYM), Vermi-Compost, Green Manure etc. may be used as per requirement of the species. To prevent diseases, bio-pesticides could be prepared (either single or mixture) from Neem (kernel, seeds & leaves), Chitrakmool, Dhatura, Cow's urine etc. Light shower after transplantation ensures establishment of seedlings.



Fig 1.2

Ashwagandha (*Withania somnifera*), also known as “Indian ginseng,” is a prominent adaptogenic herb in Ayurvedic medicine. Its roots are the primary part used for traditional and modern medicinal purposes, often ground into powder or used as extracts. The roots of *Withania somnifera*, commonly known as ashwagandha, a well-known medicinal herb in Ayurveda. The primary benefit of ashwagandha root lies in its adaptogenic properties, meaning it helps the body manage stress and maintain balance. It is widely used to reduce anxiety, improve mental clarity, and enhance overall energy levels. Additionally, ashwagandha supports immune function, promotes better sleep, and may improve stamina and physical performance. Traditionally, it is also used to boost reproductive health and vitality, making it a valuable herb for overall well-being and long-term health maintenance. The image shows the roots of *Withania somnifera* (ashwagandha), a widely used medicinal herb in Ayurveda. Its roots are valued for multiple health uses. Ashwagandha is commonly used to reduce stress and anxiety by calming the nervous system and helping the body adapt to physical and mental strain. It is also known to improve sleep quality and support brain function, including memory and concentration. Additionally, it helps boost immunity, increase energy levels, and enhance physical strength and stamina..

it is used to support reproductive health, improve vitality, and maintain overall balance in the body. Stress and Anxiety It is widely used to help the body adapt to stress and reduce levels of cortisol, the “stress hormone”. The roots shown in the image are commonly associated with Ashwagandha, a traditional medicinal herb widely used for managing stress and anxiety. Ashwagandha is known as an adaptogenic plant, meaning it helps the body cope with physical and mental stress. Its roots contain active compounds called withanolides, which may help reduce

cortisol levels, calm the nervous system, and improve emotional balance. Regular use of Ashwagandha root in herbal preparations has been traditionally linked with reduced anxiety, improved sleep quality, better concentration, and enhanced resistance to fatigue. It is commonly used in Ayurvedic medicine to support mental well-being and promote relaxation naturally

Physical Performance Supplementation is linked to improved muscle strength, endurance, and recovery for both athletes and non-athletes. The image shows dried roots of Ashwagandha, a traditional medicinal plant widely used in Ayurvedic medicine for improving physical performance and overall vitality. Ashwagandha roots are known for their adaptogenic properties, which help the body cope with stress and fatigue. Regular use may enhance muscle strength, stamina, endurance, and energy levels, making it beneficial for athletes and physically active individuals. It is also believed to support faster recovery after exercise, improve physical resilience, and reduce weakness. In addition, Ashwagandha may help improve concentration, balance hormonal functions, and promote better overall health and fitness.

Cognitive Function: It may enhance memory, attention, and overall brain function, and is being researched for potential neuroprotective effects in conditions like Alzheimer's and Parkinson's. The image appears to show dried roots of Ashwagandha, a traditional medicinal herb widely used in Ayurveda the role of

Ashwagandha is known for its potential role in supporting cognitive function and brain health. Research suggests that its bioactive compounds, called withanolides, may help improve memory, attention, mental clarity, and learning ability. It is also believed to reduce oxidative stress and support communication between nerve cells, which may enhance overall brain performance. In addition, Ashwagandha may help lower stress and anxiety levels, indirectly improving concentration, focus, and cognitive efficiency. Because of these properties, it is often used as a natural adaptogen to support mental well-being and cognitive health. **Deep Sleep Quality:** Ashwagandha is often used to treat insomnia and improve deep sleep by helping to rest and restore the nervous system.

Reproductive Health: In men, it is used to boost testosterone levels and improve sperm quality and count. The dried roots of Ashwagandha, an important herb in Ayurveda known for its potential benefits in reproductive health. Ashwagandha is traditionally used to support fertility, hormonal balance, and overall reproductive wellness in both men and women. In men, studies suggest that it may help improve sperm count, sperm motility, testosterone levels, and reproductive vitality by reducing oxidative stress and supporting endocrine function. In women, Ashwagandha may help manage stress-related hormonal imbalance, support menstrual health, and promote overall reproductive well-being. Its adaptogenic properties also help reduce stress and fatigue, which can positively influence reproductive function and sexual health. Because of these benefits, Ashwagandha is widely valued as a natural rejuvenating herb for reproductive support and vitality.

Topical Uses: Crushed root paste can be applied externally to reduce joint inflammation, boils, and painful swellings. The s dried roots of Ashwagandha, a medicinal herb widely valued in Ayurveda for its therapeutic and topical applications. Ashwagandha root paste and extracts are traditionally applied to the skin to help manage wounds, inflammation, joint pain, and skin infections. Its anti-inflammatory and antioxidant properties may support faster healing and help soothe irritated skin. In traditional practices, Ashwagandha paste is also used for reducing swelling, muscle

discomfort, and localized pain. Some herbal preparations containing Ashwagandha are applied to improve skin texture and support overall skin health by protecting against oxidative stress. Because of these beneficial properties, Ashwagandha is commonly used in natural topical remedies for skin care and .

General Tonic: Ashwagandha is a small shrub traditionally used in Ayurvedic medicine, especially in India. Its roots are the most commonly used part of the plant and are often dried, powdered, or extracted for supplements and herbal preparations. **Well-being Health:** Some studies suggest it helps manage blood sugar levels by stimulating insulin secretion. Ashwagandha roots are widely valued in traditional herbal medicine for their diverse therapeutic properties and long history of use in promoting overall health and well-being. The roots contain important bioactive compounds called withanolides, alkaloids, and flavonoids that contribute to their medicinal effects. Ashwagandha roots are commonly used as an adaptogen, helping the body manage stress, reduce anxiety, and improve mental balance. They are also known for supporting physical strength, stamina, immunity, and energy levels. In addition, the roots have antioxidant, anti-inflammatory, and rejuvenating properties that may help protect the body from fatigue and age-related disorders. Traditionally, Ashwagandha roots have been used to support cognitive function, reproductive health, sleep quality, and recovery from weakness or illness. Due to these wide-ranging health benefits, Ashwagandha roots remain one of the most important



Fig 1.3

Ashwagandha plant (*Withania somnifera*), also known as “Indian Ginseng” or “Winter Cherry”. While the roots are the most commonly used part of the plant in traditional medicine, the leaves also possess significant medicinal properties. The leaves and developing fruits of *Withania somnifera* (ashwagandha), a versatile medicinal herb widely used in traditional Ayurvedic practices. The leaves of ashwagandha are known for their significant therapeutic uses, particularly for their anti-inflammatory, antimicrobial, and analgesic properties. They are often applied externally in the form of pastes to help treat wounds, skin infections, swelling, and joint pain, providing relief from discomfort and promoting faster healing. Internally, in controlled traditional preparations, the leaves may contribute to reducing fever, managing pain, and supporting the body’s immune response. In addition, ashwagandha leaves are sometimes used to help manage stress and anxiety due to their calming effects on the nervous system, complementing the well-known adaptogenic properties of the plant’s roots. They also play a role in improving overall health by supporting detoxification processes and enhancing resistance to infections. Although the roots remain the most commonly used part for supplements and tonics, the leaves are an important component in traditional medicine systems, offering a wide range of

uses that contribute to maintaining balance, vitality, and holistic well-being.

The leaves of *Withania somnifera* (ashwagandha) are simple, ovate to elliptic in shape, with smooth (entire) margins and a slightly pointed tip. They are typically dull green in color with a soft, slightly velvety texture due to fine hairs on the surface. The leaves are arranged alternately on the stem and have a short petiole. Prominent midribs with fine lateral veins are visible, giving structural support to the leaf blade. They are medium-sized, usually about 5–10 cm long, and have a somewhat thick, leathery feel, which helps the plant adapt to dry conditions. They are usually alternate on vegetative shoots but appear opposite on floral branches. Ashwagandha leaves are traditionally used in Ayurveda to manage stress, reduce inflammation, and alleviate pain due to their potent medicinal properties. They are often applied topically as a paste for swelling or consumed as tea/juice to boost immunity, improve digestion, and combat fatigue.

The leaves of *Withania somnifera* (ashwagandha) are widely used in traditional medicine for their therapeutic properties. They are especially valued for their anti-inflammatory and pain-relieving effects, often applied as a paste on swollen joints, wounds, and skin infections to promote healing and reduce discomfort. The leaves also possess antimicrobial activity, helping to protect against bacterial and fungal infections. In some traditional practices, they are used in decoctions or extracts to help reduce fever, support immunity, and assist the body in fighting infections. Additionally, ashwagandha leaves have mild calming properties, contributing to stress reduction and relaxation. They are also sometimes used in herbal formulations for detoxification and improving overall health, although they are generally considered more potent and slightly bitter compared to the root and thus used carefully under guidance.

Inflammation and Pain: Topical application of leaf paste is used to treat painful swellings, joint pains, and skin infections. The leaves of Ashwagandha are traditionally valued for their anti-inflammatory and pain-relieving properties. They contain bioactive compounds such as withanolides, alkaloids, and flavonoids that help reduce inflammation in the body and may ease discomfort associated with joint pain, muscle soreness, arthritis, and swelling. In traditional Ayurvedic medicine, Ashwagandha leaves have been used in poultices and herbal preparations to soothe wounds, skin irritation, and localized pain. Modern studies also suggest that the leaves possess antioxidant activity, which may help protect tissues from oxidative stress and support faster recovery from inflammation-related conditions.

Stress and Anxiety: Similar to the root, the leaves contain adaptogenic properties that help the body manage stress and reduce cortisol levels. The leaves of Ashwagandha are known for their calming and adaptogenic properties, which help the body manage stress and anxiety naturally. Ashwagandha leaves contain important bioactive compounds, including withanolides and flavonoids, that may support the nervous system, reduce cortisol levels, and promote mental relaxation. In traditional Ayurvedic medicine, the leaves have been used to ease nervous tension, improve emotional balance, and support restful sleep. Their antioxidant and neuroprotective activities may also help protect brain cells from stress-related damage, contributing to improved mood, mental clarity, and overall psychological well-being.

Immune System Booster: The leaves are rich in antioxidants, including iron and vitamin C, which help in enhancing immune function. **Respiratory Support:** A decoction or tea made from the leaves is used to treat coughs. The leaves of Ashwagandha are considered beneficial for strengthening the immune system due to their rich content of withanolides, alkaloids, and antioxidant compounds. These bioactive substances help enhance the body's natural defense mechanisms by supporting immune cell activity and protecting cells from oxidative stress caused by free radicals. In traditional Ayurvedic medicine, Ashwagandha leaves have been used to improve overall vitality, increase resistance to infections, and promote faster recovery from illness. Their anti-inflammatory and antimicrobial properties may also help the body maintain better health and balance, contributing to improved immunity and general well-being.



Fig 1.4

Ashwagandha is a well-known Ayurvedic herb valued for its wide range of therapeutic properties. It is commonly used as an adaptogen, meaning it helps the body cope with stress and restore balance. Ashwagandha is an important medicinal herb widely used in traditional Ayurvedic medicine for its numerous therapeutic properties. The plant contains active compounds such as withanolides, alkaloids, saponins, and flavonoids that contribute to its antioxidant, anti-inflammatory, antimicrobial, adaptogenic, and immunomodulatory activities. Different parts of the plant, including the roots, leaves, and berries, are used to support physical strength, reduce stress and anxiety, improve memory and cognitive function, enhance reproductive health, and boost immunity. Ashwagandha is also traditionally used for relieving joint pain, fatigue, insomnia, skin disorders, and general weakness. Due to its wide range of medicinal benefits, it is considered a valuable natural remedy for promoting overall health and well-being..

Stress & Anxiety Relief: Ashwagandha helps reduce cortisol (stress hormone) levels, promoting calmness and improving mental well-being. It is often used to manage anxiety, tension, and fatigue. Ashwagandha is widely recognized for its ability to relieve stress and anxiety through its adaptogenic properties. The herb helps the body cope with physical and mental stress by regulating cortisol, the primary stress hormone, and supporting healthy nervous system function. Active compounds such as withanolides and alkaloids contribute to its calming effects, which may help reduce nervous tension, anxiety, irritability, and mental fatigue. In traditional Ayurvedic medicine, Ashwagandha has long been used to promote emotional balance, improve relaxation, and support restful sleep. Regular use may also enhance mood, mental clarity, and overall psychological well-

being, making it a valuable natural remedy for stress management.

Boosts Immunity: It strengthens the immune system by enhancing white blood cell activity and increasing the body's resistance to infections. The leaves of Ashwagandha are known to help boost immunity due to their rich concentration of withanolides, antioxidants, and other beneficial phytochemicals. These compounds support the body's natural defense system by enhancing immune cell activity and protecting tissues from oxidative stress and harmful free radicals. In traditional Ayurvedic medicine, Ashwagandha leaves have been used to improve strength, vitality, and resistance against infections and diseases. Their anti-inflammatory and antimicrobial properties may also help the body recover more effectively from illness, contributing to better overall health and a stronger immune system.

Improves Energy & Stamina: The leaves of Ashwagandha are traditionally used to improve energy, stamina, and overall physical performance. They contain bioactive compounds such as withanolides, flavonoids, and alkaloids that help reduce fatigue, enhance endurance, and support the body's ability to cope with physical stress. In Ayurvedic medicine, Ashwagandha leaves are valued as a natural rejuvenating tonic that promotes strength, vitality, and recovery from weakness or exhaustion. Their adaptogenic and antioxidant properties may also help improve oxygen utilization, support muscle function, and increase overall energy levels, making the plant beneficial for maintaining physical and mental stamina.

Supports Brain Function: Ashwagandha enhances memory, focus, and cognitive performance. It may also help in slowing age-related mental decline. The leaves of Ashwagandha are believed to support healthy brain function due to their neuroprotective and antioxidant properties. They contain bioactive compounds such as withanolides that may help protect brain cells from oxidative stress and improve communication between nerve cells. In traditional Ayurvedic medicine, Ashwagandha leaves have been used to enhance memory, concentration, mental clarity, and learning ability. Their calming effects may also help reduce stress and mental fatigue, which can positively influence cognitive performance and emotional balance. Regular use of Ashwagandha is traditionally associated with improved overall brain health and nervous system support.

Anti-inflammatory & Pain Relief: reduces inflammation and is beneficial for conditions like arthritis, joint pain, and muscle soreness. The leaves of Ashwagandha possess significant anti-inflammatory and pain-relieving properties that make them valuable in traditional medicine. They contain active compounds such as withanolides, alkaloids, and flavonoids, which help reduce inflammation, swelling, and discomfort in the body. Ashwagandha leaves are traditionally used to relieve joint pain, muscle soreness, arthritis, and inflammatory conditions by helping calm the body's inflammatory response. Their antioxidant properties also help protect tissues from oxidative damage and support faster healing and recovery. In Ayurvedic practices, leaf extracts and poultices are commonly applied externally to soothe pain, wounds, and skin irritation naturally.

Hormonal Balance: Ashwagandha supports thyroid function and helps balance hormones. In

men, it may improve testosterone levels and fertility; in women, it may help regulate cycles. The leaves of Ashwagandha are traditionally believed to support hormonal balance by helping regulate the body's stress response and endocrine system function. Bioactive compounds such as withanolides may help control cortisol levels, which can positively influence the balance of other hormones related to mood, energy, metabolism, and reproductive health. In Ayurvedic medicine, Ashwagandha leaves have been used to support thyroid function, improve vitality, and promote overall hormonal stability in both men and women. Their adaptogenic and antioxidant properties may also help reduce stress-related hormonal imbalances, contributing to improved physical health, emotional well-being, and overall body

Sugar Control : The leaves of Ashwagandha are traditionally used to help support healthy blood sugar control and metabolic balance. They contain beneficial compounds such as withanolides, flavonoids, and antioxidants that may improve insulin sensitivity and assist in regulating glucose metabolism. In Ayurvedic medicine, Ashwagandha leaves have been used as a natural remedy to help manage fatigue and weakness associated with fluctuating blood sugar levels. Their antioxidant and anti-inflammatory properties may also help protect body tissues from oxidative stress linked to high glucose levels, contributing to better overall metabolic health and well-being.

Heart Health : It may reduce cholesterol and improve overall cardiovascular health. The leaves of Ashwagandha are believed to support heart health due to their antioxidant, anti-inflammatory, and adaptogenic properties. They contain bioactive compounds such as withanolides and flavonoids that may help reduce oxidative stress, improve blood circulation, and support healthy cardiovascular function. In traditional Ayurvedic medicine, Ashwagandha leaves have been used to help manage stress and fatigue, which are important factors affecting heart health. Their calming effects may assist in maintaining balanced blood pressure and reducing stress-related strain on the heart. Additionally, the antioxidant activity of the leaves may help protect blood vessels and heart tissues from cellular damage, contributing to overall cardiovascular well-being.

Skin and Wound Care: The leaves are used as a topical paste for skin ulcers, scabies, leprosy, and to treat swellings. The leaves of Ashwagandha are traditionally used in skin and wound care because of their anti-inflammatory, antimicrobial, and antioxidant properties. The bioactive compounds present in the leaves, including withanolides and flavonoids, may help protect the skin from infections, reduce irritation, and support faster healing of wounds, cuts, and burns. In Ayurvedic medicine, Ashwagandha leaf paste and extracts are commonly applied externally to soothe skin inflammation, ulcers, and minor injuries. Their antioxidant activity may also help reduce oxidative damage to skin cells, promoting healthier skin appearance and supporting natural tissue repair and regeneration

Respiratory and Other Uses: Leaves are used to treat asthma, cough, and fever. The leaves of Ashwagandha are traditionally used to support respiratory health and treat various other health conditions in Ayurvedic medicine. Their anti-inflammatory, antimicrobial, and antioxidant properties may help relieve cough, cold, asthma, bronchitis, and throat irritation by reducing inflammation in

the respiratory tract and supporting easier breathing. Ashwagandha leaves are also valued for their ability to strengthen immunity and improve the body's resistance to infections. In addition to respiratory benefits, the leaves have been used for wound healing, pain relief, skin care, stress reduction, improving energy levels, and supporting overall physical and mental well-being. Their wide range of medicinal properties makes Ashwagandha an important herb in traditional natural medicine.



Fig 1.5

Ashwagandha fruit (berries) and roots, often used in Ayurveda, provide potent health benefits including reduced stress and anxiety, improved sleep quality, lower blood sugar, and increased physical strength. As a powerful adaptogen, it lowers cortisol levels, acts as an anti-inflammatory, and supports cognitive function, enhancing The fruits of Ashwagandha are traditionally valued in herbal medicine for their potential role in reducing stress and anxiety. They contain bioactive compounds such as withanolides, alkaloids, and antioxidants that may help support the body's natural stress response and promote emotional balance. Ashwagandha fruit is considered an adaptogenic plant material, meaning it may help the body cope with physical and mental stress more effectively. Regular use in traditional preparations is believed to calm the nervous system, reduce feelings of nervousness and fatigue, and improve overall mental well-being. Some studies on Ashwagandha plant extracts also suggest that its compounds may help lower cortisol levels, the hormone associated with stress, thereby supporting relaxation and improving mood stability. Acts as an adaptogen that lowers cortisol levels and calms the nervous system, assisting with anxiety management.

The fruits of Ashwagandha are traditionally used in herbal practices to support better sleep and relaxation. They contain natural bioactive compounds, including withanolides and antioxidants, which may help calm the nervous system and reduce mental stress that often interferes with healthy sleep patterns. Ashwagandha fruit is believed to promote restful sleep by helping the body adapt to stress and encouraging physical and mental relaxation. Traditional medicine systems value the plant for its mild sedative and rejuvenating properties, which may improve sleep quality, reduce nighttime restlessness, and support overall energy recovery. Regular use of Ashwagandha-based preparations is also associated with improved mood balance and reduced fatigue, both of which contribute to healthier and more refreshing sleep. Often referred to as “sleep berry” (*somnifera*), it helps improve sleep quality, efficiency, and total sleep time.

The fruits of Ashwagandha are valued in traditional herbal medicine for supporting

physical performance and overall vitality. They contain beneficial phytochemicals such as withanolides, flavonoids, and antioxidants that may help improve energy production, stamina, and muscle strength. Ashwagandha fruit is considered an adaptogenic plant material that may enhance the body's ability to tolerate physical stress and fatigue during exercise or strenuous activity. Traditional uses also suggest that it may support healthy blood circulation and oxygen delivery throughout the body, which can contribute to improved endurance and recovery. Regular consumption in herbal formulations is believed to reduce weakness, support muscle function, and promote overall physical fitness and body balance. Boosts Physical Performance Studies show it increases muscle strength, cardiorespiratory endurance, and muscle size during training.

The fruits of Ashwagandha are cognitive function and mental well-being. They contain bioactive compounds such as withanolides, alkaloids, and antioxidants that may help protect brain cells from oxidative stress and support healthy nervous system activity. Ashwagandha fruit is believed to enhance memory, concentration, learning ability, and mental clarity by promoting balanced brain function and reducing stress-related mental fatigue. Traditional herbal practices also associate the plant with improved focus and emotional stability, which may contribute to better cognitive performance. Regular use of Ashwagandha-based preparations is thought to support overall brain health, mental alertness, and long-term cognitive wellness. Cognitive Function Enhances memory, attention and mental clarity, with potential neurodegenerative conditions.

The fruits of Ashwagandha are valued for their potential role in supporting healthy blood sugar balance and strengthening the immune system. They contain natural compounds such as withanolides, flavonoids, and antioxidants that may help regulate metabolic activity and reduce oxidative stress in the body. Traditional herbal medicine suggests that Ashwagandha fruit may assist in maintaining stable blood glucose levels by supporting insulin function and improving the body's response to stress, which can influence sugar metabolism. In addition, its antioxidant and adaptogenic properties are believed to enhance immune defense by supporting the activity of immune cells and improving the body's resistance to infections and fatigue. Regular use of Ashwagandha-based preparations is associated with improved overall wellness, better energy balance, and stronger natural immunity. Blood Sugar and Immune System Helps reduce blood sugar levels and supports the immune system by increasing natural killer cell activity.

The fruits of Ashwagandha are used in herbal medicine to support fertility and sexual health. They contain bioactive compounds such as withanolides, alkaloids, and antioxidants that may help improve overall reproductive wellness and vitality. Ashwagandha fruit is believed to support hormonal balance, enhance energy levels, and reduce stress, which can positively influence reproductive function in both men and women. Traditional practices also associate the plant with improved stamina, libido, and reproductive strength. Its adaptogenic and antioxidant properties may help protect reproductive tissues from oxidative stress and support healthy circulation and endurance. Regular use of Ashwagandha-based preparations is traditionally linked with improved vitality, physical performance, and overall sexual well-being. Fertility and Sexual Health Improves sperm

count and motility in men and may help with erectile dysfunction

OBSERVATION

Sr No	Common names	Plant of use	Method Preparation	Aliment Treated	Observation effect
1	Ashwagandha	Root	Root dried powder mixed with warm milk	Weakness Fatigue	Increase strength and energy
2	Ashwagandha	Root	Root past applied externally	Join pain Swelling	Reduce pain and inflammation
3	Ashwagandha	Leaves	Leaf paste applied externally	Wounds boils	Promote faster healing
4	Ashwagandha	Leaves	Leaf decoction	Fever infection	Help reduce fever
5	Ashwagandha	Fruit	Consumed used topically in paste	Skin ulcers tumors carbuncle flavouring	The manage stress reduce anxiety
6	Ashwagandha	Seed	Crushed seed applied	Skin infection	Antimicrobial action
7	Ashwagandha	Root Bark	Decoction	Insomnia	Promotes sound sleep

RESULT AND DIUCUSSION

RESULT:-

Withnina somnifera (Ashwagandha), widely used in traditional medicine, acts as an adaptogen, anti-inflammatory, and tonic to treat stress, anxiety, rheumatism, skin ailments, and exhaustion. Tribal and traditional users commonly apply the root paste for boils, use leaf extracts for fever, and consume the root to boost immunity and reproductive health. A total of 86 plant species belonging to 75 genera and 45 families were reported with ethnomedicinal uses. In terms of the number of medicinal plant species, Acanthaceae (6 genera and 7 species, 8% of total collected plants) and Cucurbitaceae (5 species) are dominant families. Among the different plant parts used for the preparation of medicine, the leaves were most frequently used for the treatment of diseases. This critical review covers the *W. somnifera* ethnography, phytochemistry, and ethnomedicinal usage to demonstrate the use of the plant in Africa and elsewhere to prevent or alleviate several pathophysiological conditions, including cardiovascular, neurodegenerative, reproductive impotence, as well as other chronic diseases.

Adaptogenic & Anti-Stress: Acting as potent anti-stress agent, it reduces serum cortisol levels, helping the body manage stress and improving sleep quality. *Withania somnifera* is widely recognized for its adaptogenic and anti-stress properties, which help the body cope with physical, emotional, and environmental stress. Traditional tribal communities and herbal medicine practitioners use the plant to reduce anxiety, mental fatigue, nervous exhaustion, and weakness. The bioactive compounds present in Ashwagandha, particularly withanolides and alkaloids, are believed to support the nervous system and regulate stress-related hormonal activity, including cortisol levels. Regular use of Ashwagandha preparations is associated with improved emotional balance, enhanced relaxation, better sleep quality, and increased resistance to stress. Its adaptogenic nature also helps maintain energy, stamina, and overall mental well-being during periods of physical or psychological strain.

It as significant anti-inflammatory properties, effective in treating rheumatism and arthritis. *Withania somnifera* is traditionally known for its anti-inflammatory and anti-arthritic properties and has been widely used by tribal communities and herbal practitioners to relieve pain and inflammation. Different parts of the plant, especially the roots and leaves, are used in traditional remedies for treating arthritis, joint swelling, muscle stiffness, rheumatism, and inflammatory disorders. The plant contains bioactive compounds such as withanolides, flavonoids, and alkaloids that may help reduce inflammatory responses and protect tissues from oxidative damage. Regular use of Ashwagandha-based preparations is believed to improve joint mobility, reduce discomfort, and support muscle and bone health. Its natural anti-inflammatory activity contributes to overall physical wellness and makes it an important medicinal plant in traditional healing systems.

Neurological & Cognitive Health: It demonstrates potential in managing Parkinson's and Alzheimer's diseases by protecting neurons from oxidative damage and boosting antioxidant activity. *Withania somnifera* is traditionally valued for its beneficial effects on neurological and cognitive health. Tribal communities and herbal medicine practitioners use the plant to improve memory, concentration,

learning ability, and overall mental performance. The bioactive compounds present in Ashwagandha, especially withanolides and antioxidants, are believed to protect nerve cells from oxidative stress and support healthy brain function. Traditional remedies use the plant to reduce disorders, and age-related cognitive decline. Ashwagandha is also considered a calming tonic that supports emotional balance, better sleep, and improved nervous system function. Regular use of Ashwagandha-based preparations is associated enhanced mental clarity, cognitive wellness, neurological health.

immuno-modulatory & Anti-cancer Traditional Uses: Tribal communities often use root powders or leaf extracts to treat chronic fevers, insomnia, coughs, and as a general body It studies possesses immuno-modulatory properties and has been shown to slow down tumor growth in pre-clinical studies. *Withania somnifera* is traditionally recognized for its immuno-modulatory and potential anti-cancer properties in tribal and herbal medicine systems. Tribal communities use different parts of the plant to strengthen the body's natural defense mechanisms, improve resistance to infections, and support overall health during weakness and illness. The plant contains bioactive compounds such as withanolides, alkaloids, and antioxidants that may help regulate immune responses and protect cells from oxidative stress. Traditional uses also associate Ashwagandha with the management of abnormal growths, inflammation, and chronic diseases due to its rejuvenating and detoxifying effects. Modern scientific studies have further explored its potential role in inhibiting the growth of certain cancer cells and supporting recovery during severe illness. These traditional and scientific observations highlight the importance of Ashwagandha as a valuable medicinal plant for immune support and overall wellness.

Physical Performance & Strength: It increases muscle strength, cardiorespiratory endurance, and muscle size while reducing post short-term use, though it may cause side effects like digestive upset in higher doses. *Withania somnifera* is traditionally valued for improving physical performance, strength, and overall vitality. Tribal communities and herbal practitioners use the plant as a rejuvenating tonic to increase stamina, energy, endurance, and muscle strength during physical work and recovery from weakness. The plant contains beneficial bioactive compounds such as withanolides, alkaloids, and antioxidants that may help the body adapt to physical stress and reduce fatigue. Traditional remedies prepared from Ashwagandha roots, leaves, and fruits are believed to support healthy muscle function, improve body balance, and enhance recovery after exertion. Regular use of Ashwagandha-based preparations is also associated with improved physical fitness, better resistance to exhaustion, and overall strengthening of the body and nervous system.

DISCUSSION:-

Withnina somnifera (Ashwagandha), widely used in traditional medicine, acts as an adaptogen, anti-inflammatory, and tonic to treat stress, anxiety, rheumatism, skin ailments, and exhaustion. Tribal and traditional users commonly apply the root paste for boils, use leaf extracts for fever, and consume the root to boost immunity and reproductive health. *Withnina somnifera* (L.) Dunal, commonly called Ashwagandha or "Indian ginseng," has been used for over 3,000 years in Ayurveda and Unani medicine. Many of these traditional practices overlap with tribal/ethnomedicinal knowledge, especially in India's arid regions where the plant grows wild.

Acting as a Potent anti-stress agent, it reduces serum cortisol levels, helping the body manage stress and improving sleep quality. Today's lifestyle is particularly frustrating owing to the hectic schedules of everyone from school-age children to the elderly. As a result, there is a sizable market for pharmaceuticals that treat stress and psychosis globally. Stress is a major contributor to a wide range of illnesses, from psychological conditions to endocrine problems including, diabetes mellitus, hypothyroidism, male sexual dysfunction, hypertension, ulcerative colitis, peptic ulcer, etc. Chronic stress may have extremely negative impacts on our health, but acute stress can be exhilarating and keep us attentive and active¹⁸. The plant, *Withnina somnifera* shows activities like Gamma-aminobutyric acid (GABA), which may contribute for the anti-anxiety and anti-stress effects of herbs^{19,5}. Neuronal excessive activity may lead to insomnia and restlessness; however, GABA inhibits a large number of brain nerve cells that gets activated. The function of GABA is to lessen the activity of neurons and to inhibit the nerve cells from over activating, which further generates a soothing impact. It uplift mood, induce sleep and reduce stress and anxiety²⁰.

The anti-stress activities of WS have demonstrated to be highly efficient in boosting stamina (physical endurance) The obtainable documentation by ancient Ayurvedic literature explained that the *Withnina somnifera* is a potent regenerative herb and a best rasayana. In Ayurveda the whole plant is medically valuable, but roots are commonly used in formulations. Various literatures revealed that *Withnina somnifera* contains plenty of pharmacologically valuable phytochemicals that attributes the pharmacological activity of the herb. The withanolides are the vital chemical constituents and around 138 withanolides are identified. Various clinical studies underlined its efficiency of this herb to treat against number of diseases. It acts as sedative, diuretic, generally respected for increasing energy anti-inflammatory, anti-stress agent endurance, acts as an-adaptogen that exerts a strong immunostimulatory. Ashwagandha is taken for treating ulcers, cold coughs, emaciation, epilepsy, diabetes, conjunctivitis, insomnia, senile dementia, leprosy, Parkinson's disease, nervous system disorders, epilepsy, rheumatism, arthritis, intestinal infections, bronchitis, asthma, a suppressant in HIV/AIDS patients.

Anti-inflammatory & Anti-arthritis: It Has significant anti-inflammatory properties, effective in treating rheumatism and arthritis. Some of the chemicals found in *Withnina somnifera* are powerful antioxidants. Studies conducted on rats' brains showed the herb produced an increase in the levels of three natural antioxidants- superoxide dismutase, catalase and glutathione peroxidase⁴⁴. These findings are consistent with the therapeutic use of *Withnina somnifera* as an Ayurvedic rasayana. The antioxidant effect of active principles of *Withnina somnifera* root may explain the reported anti-stress, cognition-facilitation, anti-inflammatory and anti-aging effects produced by them in experimental animals, and in clinical situations. Various parts of the plant, especially the roots and leaves, contain bioactive compounds such as withanolides, alkaloids, saponins, and flavonoids that contribute to its therapeutic effects. These compounds are believed to reduce inflammation by inhibiting inflammatory mediators and minimizing oxidative stress in body tissues.

Traditional healers and tribal communities commonly use Ashwagandha preparations to manage arthritis, rheumatism, joint swelling, muscle pain, and stiffness. Scientific studies have also reported that Ashwagandha may help decrease pain, improve joint mobility, and protect cartilage and connective tissues from degeneration. Its antioxidant activity further supports tissue repair and reduces cellular damage associated with chronic inflammatory disorders. Due to these combined effects, Ashwagandha is considered an important medicinal plant for supporting musculoskeletal health and improving the quality of life in individuals suffering from inflammatory and arthritic conditions.

Anti-carcinogenic activity: Ashwagandha is reported to have anti-carcinogenic effects. Research on animal cell cultures has shown that the herb decreases the levels of the nuclear factor kappa B, suppresses the intercellular tumor necrosis factor, and potentiates apoptotic signaling in cancerous cell lines⁴⁶. One of the most exciting of the possible uses of Ashwagandha is its capacity to fight cancers by reducing tumor size⁴⁷. To investigate its use in treating various forms of cancer, the antitumor effects of *Withnina somnifera* have been studied by researchers. In one study, the herb was evaluated for its anti-tumor effect in urethane-induced lung tumors in adult male mice. Following administration of Ashwagandha over a period of seven months, the histological appearance of lungs of animals which received the herb was similar to those observed in the lungs of control animals. The plant contains several biologically active compounds, particularly withanolides such as withaferin A, which are reported to possess antioxidant, anti-inflammatory, and cytotoxic properties. These compounds may help inhibit the growth and spread of abnormal cells by inducing apoptosis (programmed cell death), suppressing tumor cell proliferation, and reducing oxidative stress that contributes to cellular damage. Experimental studies have also suggested that Ashwagandha extracts may interfere with cancer cell signaling pathways and support immune function, thereby helping the body defend against disease progression. In traditional herbal systems, the plant has long been used as a rejuvenating and restorative remedy to improve strength.

CONCLUSION

CONCLUSION

Withnina somnifera, best known as ashwagandha has been used for centuries for the treatment of vivid health disorders. Multiple health benefits featured in this herbal supplement makes it as a perfect rejuvenator of physical and psychological health. As per research, this medicinal herb is mainly found in the regions of North America and India. Powerful antioxidant compounds enriched in this herb scavenges free radicals and reduces aging impact on person. Apart from consuming this extract, diet taken by person plays an important role in increasing the level of antioxidants in body. In order to obtain good level of antioxidants, it is advised to include surplus amount of fruits and vegetables in diet. Apples, berries, onions and carrots are some among the top listed food items enriched with antioxidant compounds. The use of herbal drug is increasing worldwide as they have fewer side effects as compared with synthetic drugs. Ayurveda claims therapeutic potentials of various plants.

A lot of work has been done on this multipurpose drug yielding plant till now. But all this information is fragmented therefore, the present review has been an attempt to compile this available information in a comprehensive manner. An extensive research has been done on this plant in past three decades but still there is an urgent need to carry out investigations on the biological activities, efficacies and modes of action of this traditional drug. In India, three species of the genus *Withnina* are found, *Withnina somnifera*, *Withnina ashwagandha* and *Withnina coagulant*. Withanolides are the principal compounds found in all the three species, there are some withanolides specific to each other. Withaferin A is an important phytochemical found in *W. somnifera* and *Withnina ashwagandha*, whereas, coagulin L has been found in major amounts in *Withnina coagulants*. A unique thio-dimer of withanolide named Ashwagandhanolide has been found in *Withnina somnifera*. The plant has been used as an antioxidant, adaptogen, aphrodisiac, liver tonic, anti-inflammatory agent, anticancer, central nervous system depressant, hepatoprotective and astringent and more recently as an antibacterial, antihyperglycaemic, hypolipidaemic and antitumor, as well as to treat ulcers, senile dementia, Parkinson's and Alzheimer's. It had the greater therapeutic value overall.

The variety of activities reported for the extracts, fractions and withanolides isolated from this wonder medicinal plant provide promising evidence for future research. Withanolides could achieve an important place in the world of modern drugs. Isolation on a large scale, chemical transformations and synthesis of the active compounds will definitely enhance their pharmacological value. The pharmacophores of various pharmacologically active withanolides have not yet been identified. All these advantages prove the significance of *W. somnifera* in natural product research. Despite having immense medicinal properties a multipronged strategy is required for making Ashwagandha varieties more competitive. There is a need to augment the pharmacological properties by selecting and improving chemotypes producing prodigious amounts of the desired withanolide. The use of herbal drug is increasing worldwide as they have fewer side effects.

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